sensation comes from millions of microscopic sensory receptors located all over the skin and associated with the general sensations of contact, pressure, heat, cold and pain. Some parts of the body, such as the finger tips have a large number of these receptors, making them more sensitive. Some of the sensory receptors present in the skin (Figure 10.21) are:

- **Tactile merkel disc** is light touch receptor lying in the deeper layer of epidermis.
- **Hair follicle receptors** are light touch receptors lying around the hair follicles.
- **Meissner's corpuscles** are small light pressure receptors found just beneath the epidermis in the dermal papillae. They are numerous in hairless skin areas such as finger tips and soles of the feet.
- **Pacinian corpuscles** are the large egg shaped receptors found scattered deep in the dermis and monitoring vibration due to pressure. It allows to detect different textures, temperature, hardness and pain
- **Ruffini endings** which lie in the dermis responds to continuous pressure.
- **Krause end bulbs** are thermoreceptors that sense temperature.

Melanocytes are the cells responsible for producing the skin pigment, melanin, which gives skin its colour and protects it from the sun's UV rays. Vitiligo (Leucoderma) is a condition in which the melanin pigment is lost from areas of the skin, causing white patches, often with no clear cause. Vitiligo is not contagious. It can affect people of any age, gender, or ethnic group. The patches appear when melanocytes fails to synthesis melanin pigment.

Summary

Neural system coordinates and integrates the functions of all organs and responds to changes in the internal and external environments.

Neural system includes two types of cells neurons and neuroglia. Neuron forms the structural and functional unit of the neural system.

CNS includes brain and spinal cord. The major parts of the brain are the cerebrum, diencephalon, cerebellum and the brain stem. The brain is protected by the cranium and meninges. CSF provides mechanical protection and nutrients to the CNS.

The spinal cord is the continuation of the medulla oblongata and ends at the second lumbar vertebra as conus medullaris. The components involved in reflex action are called reflex arc.

There are 12 pairs of cranial nerves which arise from the brain and 31 pairs of spinal nerves from the spinal cord make the PNS. The PNS is further divided into Somatic Neural system and Autonomic Neural system. The SNS operates under conscious control. The ANS usually operates unconsciously.

The motor portion of ANS has two major divisions: sympathetic and parasympathetic.

Sensation is the conscious or subconscious awareness of changes in the external or internal environment. Four events namely stimulation, transduction, generation of impulses and integration occurs typically for a sensation to take place.

Simple receptors are associated with the general senses such as somatic senses (skin); complex receptors are associated with the special senses such as smell, taste, vision, hearing and equilibrium.

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